

# 1D CNN — Two-Hand Dynamic Gesture Run Report

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## Data paths

|                      |                                                                   |
|----------------------|-------------------------------------------------------------------|
| Training data path 1 | /home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic · 35 trials   |
| Training data path 2 | /home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic · 35 trials  |
| Training data path 3 | /home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic · 5 trials |
| Training data path 4 | /home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic · 35 trials |
| Training data path 5 | /home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic · 35 trials   |
| Total training paths | 5 · 145 trials total                                              |
| Batch-test data root | /home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic             |
| Saved model path     | saved_models/cnn_model_2026-05-06_16-26-14_e88a816fd825.keras     |
| Experiment log       | saved_models/experiment_results.csv                               |

## Sensor selection

|          |                                                |
|----------|------------------------------------------------|
| Hands    | left, right                                    |
| Segments | wrist, palm, thumb, index, middle, ring, pinky |
| Features | YPR, Accel, Flex                               |
| Channels | 176 (sequence length 90)                       |

## Preprocessing & split

|                  |                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------|
| Resample steps   | 90                                                                                                          |
| Butterworth      | on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz                                                                  |
| Normalisation    | minmax                                                                                                      |
| Test size / seed | 0.1 · random_state=42 · stratify=True                                                                       |
| Sample counts    | total=145 · train pool=130 · train+aug=780 · hold-out=15                                                    |
| Classes          | 7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll |

## Augmentation (training only)

|                      |                                          |
|----------------------|------------------------------------------|
| Enabled              | yes                                      |
| Copies per sample    | 5                                        |
| Random seed          | 42                                       |
| Active augmentations | amplitude_scale, baseline_offset         |
| • amplitude_scale    | apply_prob=0.5, scale_range=(0.95, 1.05) |
| • baseline_offset    | apply_prob=0.5, offset_std_ratio=0.02    |

## Model architecture & training

|                                 |                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Architecture                    | Conv1D(32, k=4, ReLU) → MaxPool(2) → Conv1D(64, k=4, ReLU) → MaxPool(2) → Flatten → Dense(64, ReLU) → Dropout(0.3) → Dense(softmax) |
| Optimizer / loss                | adam · sparse_categorical_crossentropy                                                                                              |
| Input shape                     | (90, 176)                                                                                                                           |
| Trainable params                | 113,255                                                                                                                             |
| Epochs · batch size · val split | 15 · 16 · 0.2                                                                                                                       |

## Cross-validation

|                    |                 |
|--------------------|-----------------|
| Folds · shuffle    | 5 · True        |
| Mean ± std val acc | 0.9154 ± 0.0449 |
| Best fold          | 2               |

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 624              | 26  | 0.9231  | 0.9231   | 0.9231    |
| 2    | 624              | 26  | 0.9615  | 0.9615   | 0.9615    |
| 3    | 624              | 26  | 0.9615  | 0.9615   | 0.9615    |
| 4    | 624              | 26  | 0.8846  | 0.9615   | 0.8846    |
| 5    | 624              | 26  | 0.8462  | 0.8846   | 0.8462    |

## Final training & hold-out test

|                                |                 |
|--------------------------------|-----------------|
| Final train acc / loss         | 0.9583 · 0.1181 |
| Final val acc / loss           | 0.8974 · 0.2263 |
| Best val acc / lowest val loss | 0.9359 · 0.1351 |
| Hold-out test acc / loss       | 0.8667 · 0.3776 |

## Per-class hold-out metrics

| Class          | Precision | Recall | F1    | Support |
|----------------|-----------|--------|-------|---------|
| Double_Lower   | 1.000     | 1.000  | 1.000 | 2       |
| Double_Nothing | 1.000     | 0.500  | 0.667 | 2       |

|                      |       |       |       |    |
|----------------------|-------|-------|-------|----|
| Double_Pistol_Recoil | 0.667 | 1.000 | 0.800 | 2  |
| Double_Raise         | 1.000 | 1.000 | 1.000 | 2  |
| Double_Wiggle        | 1.000 | 0.500 | 0.667 | 2  |
| Double_cmere         | 1.000 | 1.000 | 1.000 | 3  |
| Drum_Roll            | 0.667 | 1.000 | 0.800 | 2  |
| macro avg            | 0.905 | 0.857 | 0.848 | 15 |
| weighted avg         | 0.911 | 0.867 | 0.858 | 15 |

Batch test (NewTestData)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 4       | 5     | 0.800    |
| Double_Pistol_Recoil | 5       | 5     | 1.000    |
| Double_Raise         | 0       | 5     | 0.000    |
| Double_Wiggle        | 1       | 4     | 0.250    |
| Double_cmere         | 5       | 5     | 1.000    |
| Overall              | 15      | 24    | 0.625    |